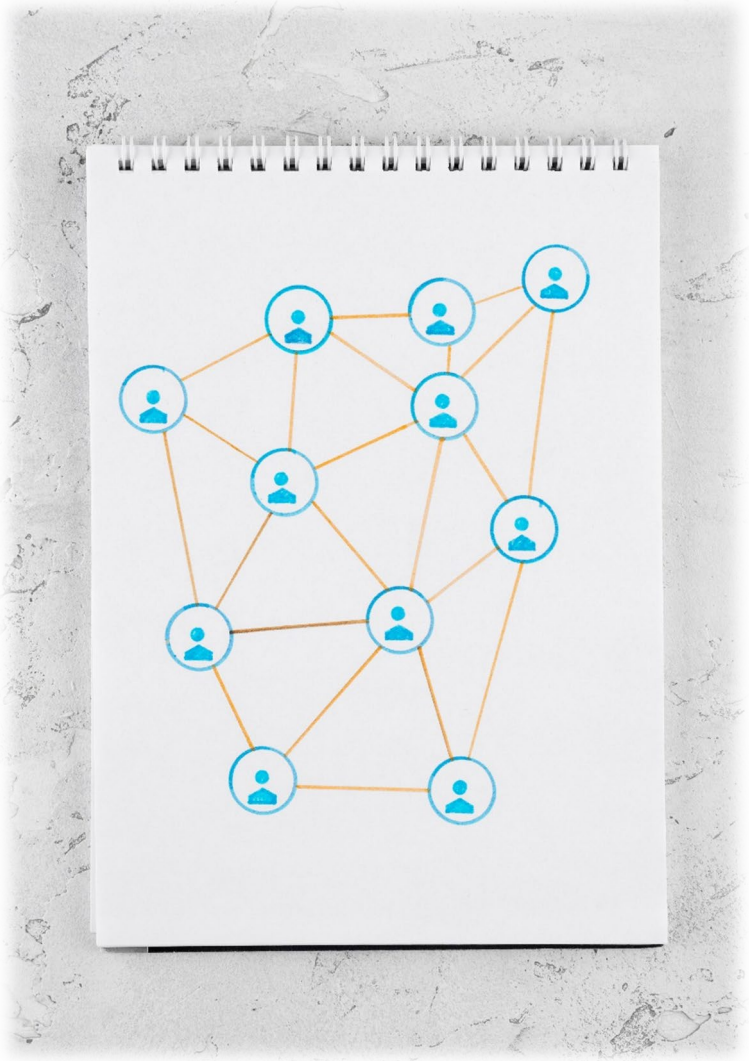


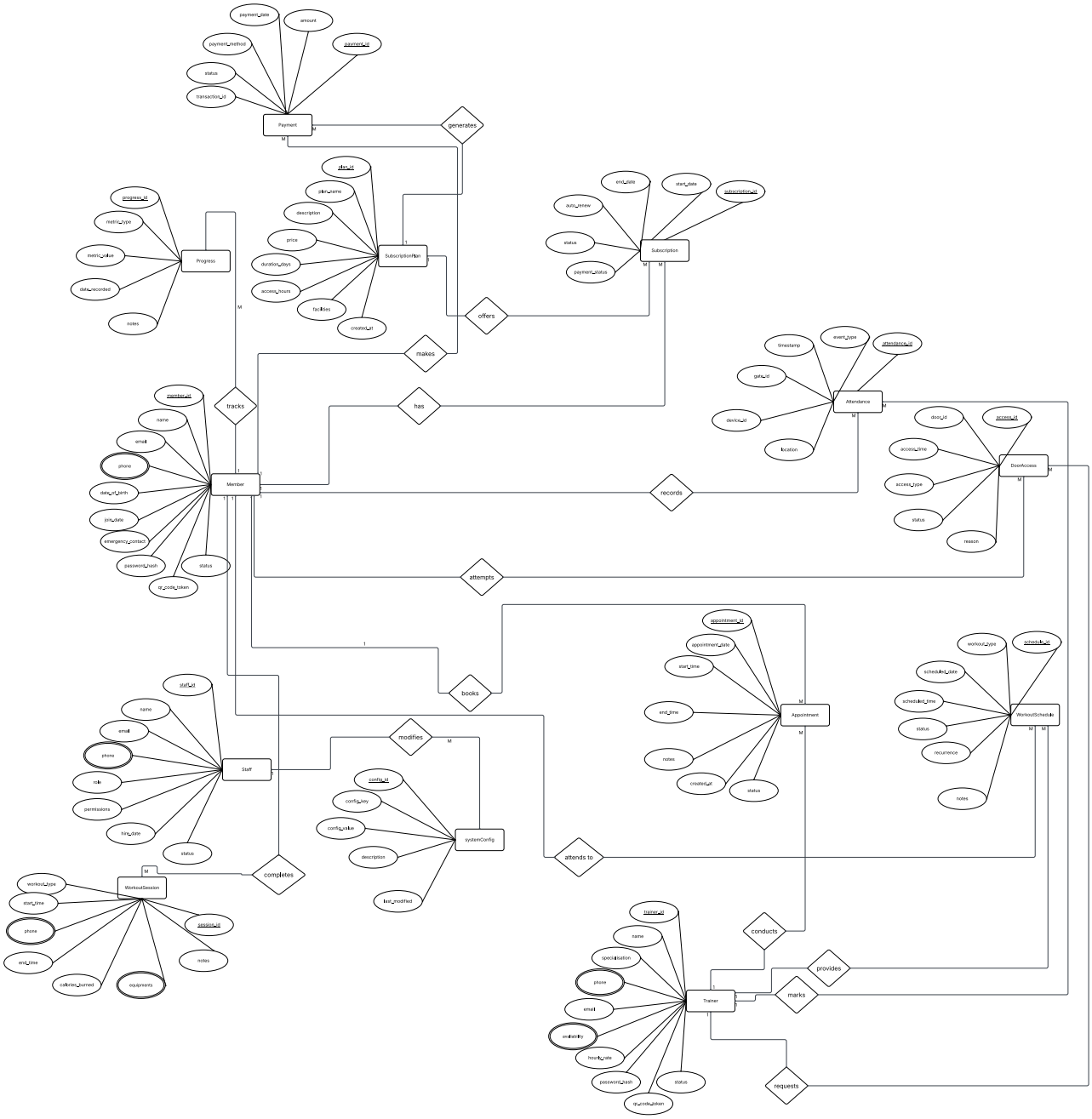
ER, DATA DICTIONARY AND SEQUENCE DIAGRAMS

POWER WORLD GYM - KIRIBATHGODA

IM/2022/001 – A.M.K.C.S.B. Abeykoon

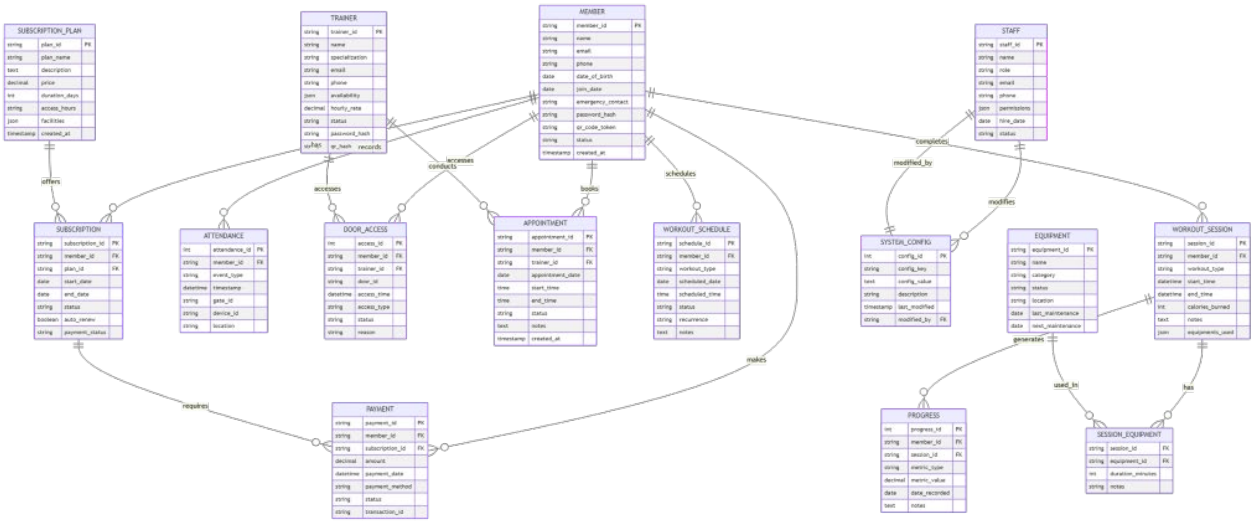
Entity-Relationship Diagram

The ER diagram graphically represents the structure of the database system, containing entities such as Member, Trainer, Staff, Subscription, Payment, and Attendance, among others, along with their respective relationships. It organizes the data in a structured way to support the smooth operation of the gym management system, including managing memberships, tracking payments, scheduling sessions, and monitoring attendance.



Relational Data Model

The ER diagram forms the basis of the relational model, converting entities and relationships into tables with fields and keys for efficient data management.



Complete Data Dictionary

A data dictionary is a detailed description of the structure of a database. It defines each table, its fields, data types, constraints, and relationships. By using the data dictionary, the database can be accurately understood, maintained, and rebuilt if needed.

MEMBER Table

Description: Stores all gym member information and authentication details

Column Name	Data Type	Constraints	Description	Example
member_id	VARCHAR(20)	PRIMARY KEY	Unique member identifier	MEM001
name	VARCHAR(100)	NOT NULL	Full name of member	John Smith
email	VARCHAR(100)	UNIQUE, NOT NULL	Contact email	john.smith@email.com
phone	VARCHAR(15)	NULL	Phone number	+94123456789
date_of_birth	DATE	NULL	Birth date	1990-05-15
join_date	DATE	NOT NULL	Membership start date	2024-01-15
emergency_contact	VARCHAR(100)	NULL	Emergency contact	Jane Smith - 0771234567

password_hash	VARCHAR(255)	NULL	Hashed password	bcrypt hash
qr_code_token	VARCHAR(100)	UNIQUE	QR code token	qr_abc123def456
status	ENUM('active','inactive','suspended')	DEFAULT 'active'	Status	active
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation timestamp	2024-01-15 09:30:00

TRAINER Table

Description: Stores trainer information and authentication details

Column Name	Data Type	Constraints	Description	Example
trainer_id	VARCHAR(10)	PRIMARY KEY	Unique trainer identifier	TRN001
name	VARCHAR(100)	NOT NULL	Full name	Sarah Johnson
specialization	VARCHAR(100)	NULL	Specialization	Weightlifting, Yoga
email	VARCHAR(100)	UNIQUE, NOT NULL	Contact email	sarah.j@email.com

phone	VARCHAR(15)	NULL	Phone number	+94123456788
availability	JSON	NULL	Availability schedule	{"mon": ["09:00-17:00"]}
hourly_rate	DECIMAL(8,2)	NULL	Hourly rate	2500.00
status	ENUM('active','inactive')	DEFAULT 'active'	Status	active
password_hash	VARCHAR(255)	NULL	Hashed password	bcrypt hash
qr_hash	VARCHAR(100)	UNIQUE	QR hash for access	qr_trainer_xyz789

STAFF Table

Description: Stores staff members information and roles

Column Name	Data Type	Constraints	Description	Example
staff_id	VARCHAR(10)	PRIMARY KEY	Unique staff identifier	STF001
name	VARCHAR(100)	NOT NULL	Full name	Mike Davis
role	ENUM('admin','instructor','receptionist','manager')	NOT NULL	Staff role	admin
email	VARCHAR(100)	UNIQUE,	Contact email	mike.d@email.com

		NOT NULL		
phone	VARCHAR(15)	NULL	Phone number	+94123456787
permissions	JSON	NULL	Permissions	{"can_manage_members": true}
hire_date	DATE	NOT NULL	Hire date	2023-06-01
status	ENUM('active','inactive')	DEFAULT 'active'	Status	active

SUBSCRIPTION_PLAN Table

Description: Stores available subscription plans and pricing

Column Name	Data Type	Constraints	Description	Example
plan_id	VARCHAR(10)	PRIMARY KEY	Unique plan identifier	PLAN001
plan_name	VARCHAR(50)	NOT NULL	Plan name	Basic Monthly
description	TEXT	NULL	Plan description	Access to gym equipment
price	DECIMAL(10,2)	NOT NULL	Price	3000.00

duration_days	INT	NOT NULL	Duration in days	30
access_hours	VARCHAR(100)	NULL	Access hours	06:00-22:00
facilities	JSON	NULL	Facilities access	{"gym": true, "pool": false}
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation time	2024-01-01 00:00:00

SUBSCRIPTION Table

Description: Stores member subscription records and status

Column Name	Data Type	Constraints	Description	Example
subscription_id	VARCHAR(20)	PRIMARY KEY	Unique subscription ID	SUB00120240115
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
plan_id	VARCHAR(10)	FOREIGN KEY	Plan reference	PLAN001
start_date	DATE	NOT NULL	Start date	2024-01-15
end_date	DATE	NOT NULL	End date	2024-02-14

status	ENUM('active','expired','cancelled')	DEFAULT 'active'	Status	active
auto_renew	BOOLEAN	DEFAULT TRUE	Auto-renewal	true
payment_status	ENUM('paid','pending','failed')	DEFAULT 'pending'	Payment status	paid

ATTENDANCE Table

Description: Records member check-in and check-out timestamps

Column Name	Data Type	Constraints	Description	Example
attendance_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique attendance ID	1
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
event_type	ENUM('IN','OUT')	NOT NULL	Event type	IN
timestamp	DATETIME	NOT NULL	Event time	2024-01-15 08:30:00
gate_id	VARCHAR(10)	NULL	Gate identifier	GATE01
device_id	VARCHAR(20)	NULL	Device identifier	SCANNER01

location	VARCHAR(50)	NULL	Location	Main Entrance

DOOR_ACCESS Table

Description: Records door access attempts for members and trainers

Column Name	Data Type	Constraints	Description	Example
access_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique access ID	1
member_id	VARCHAR(20)	FOREIGN KEY, NULL	Member reference	MEM001
trainer_id	VARCHAR(10)	FOREIGN KEY, NULL	Trainer reference	TRN001
door_id	VARCHAR(10)	NOT NULL	Door identifier	DOOR01
access_time	DATETIME	NOT NULL	Access time	2024-01-15 08:30:00
access_type	ENUM('entry','exit')	NOT NULL	Access type	entry
status	ENUM('granted','denied')	NOT NULL	Access status	granted
reason	VARCHAR(200)	NULL	Reason for denial	Subscription expired

APPOINTMENT Table

Description: Stores trainer appointments and scheduling information

Column Name	Data Type	Constraints	Description	Example
appointment_id	VARCHAR(20)	PRIMARY KEY	Unique appointment ID	APP00120240115
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
trainer_id	VARCHAR(10)	FOREIGN KEY	Trainer reference	TRN001
appointment_date	DATE	NOT NULL	Appointment date	2024-01-15
start_time	TIME	NOT NULL	Start time	09:00:00
end_time	TIME	NOT NULL	End time	10:00:00
status	ENUM('scheduled','completed','cancelled','no-show')	DEFAULT 'scheduled'	Status	scheduled
notes	TEXT	NULL	Notes	Initial consultation
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation time	2024-01-10 14:00:00

WORKOUT_SCHEDULE Table

Description: Stores scheduled workouts for members

Column Name	Data Type	Constraints	Description	Example
schedule_id	VARCHAR(20)	PRIMARY KEY	Unique schedule ID	WS00120240115
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
workout_type	VARCHAR(50)	NOT NULL	Workout type	Cardio
scheduled_date	DATE	NOT NULL	Scheduled date	2024-01-15
scheduled_time	TIME	NULL	Scheduled time	08:00:00
status	ENUM('pending','completed','skipped')	DEFAULT 'pending'	Status	pending
recurrence	ENUM('none','daily','weekly','monthly')	DEFAULT 'none'	Recurrence	none
notes	TEXT	NULL	Notes	Morning run

WORKOUT_SESSION Table

Description: Records completed workout sessions with equipment usage

Column Name	Data Type	Constraints	Description	Example
session_id	VARCHAR(20)	PRIMARY KEY	Unique session ID	SESS00120240115
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
workout_type	VARCHAR(50)	NOT NULL	Workout type	Strength Training
start_time	DATETIME	NOT NULL	Start time	2024-01-15 08:30:00
end_time	DATETIME	NULL	End time	2024-01-15 09:30:00
calories_burned	INT	NULL	Calories burned	450
notes	TEXT	NULL	Notes	Chest and triceps
equipments_used	JSON	NULL	Equipment used	[{"equipment": "Bench Press", "duration": 20}]

PROGRESS Table

Description: Tracks member progress metrics and measurements

Column Name	Data Type	Constraints	Description	Example
progress_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique progress ID	1
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
session_id	VARCHAR(20)	FOREIGN KEY, NULL	Session reference	SESS00120240115
metric_type	VARCHAR(50)	NOT NULL	Metric type	weight, body_fat
metric_value	DECIMAL(10, 2)	NOT NULL	Metric value	75.5
date_recorded	DATE	NOT NULL	Date recorded	2024-01-15
notes	TEXT	NULL	Notes	After workout

PAYMENT Table

Description: Stores payment records and transaction details

Column Name	Data Type	Constraints	Description	Example
payment_id	VARCHAR(20)	PRIMARY KEY	Unique payment ID	PAY00120240115
member_id	VARCHAR(20)	FOREIGN KEY	Member reference	MEM001
subscription_id	VARCHAR(20)	FOREIGN KEY	Subscription reference	SUB00120240115
amount	DECIMAL(10,2)	NOT NULL	Payment amount	3000.00
payment_date	DATETIME	NOT NULL	Payment date	2024-01-15 10:00:00
payment_method	VARCHAR(50)	NOT NULL	Payment method	credit_card
status	ENUM('success','failed','pending')	NOT NULL	Payment status	success
transaction_id	VARCHAR(100)	NULL	Transaction ID	txn_123456

SYSTEM_CONFIG Table

Description: Stores system configuration settings and parameters

Column Name	Data Type	Constraints	Description	Example
config_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique config ID	1
config_key	VARCHAR(100)	UNIQUE, NOT NULL	Configuration key	gym.opening.hours
config_value	TEXT	NOT NULL	Configuration value	06:00-22:00
description	VARCHAR(200)	NULL	Description	Gym opening hours
last_modified	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last modified	2024-01-15 12:00:00
modified_by	VARCHAR(10)	FOREIGN KEY	Staff who modified	STF001

EQUIPMENT Table

Description: Stores gym equipment information and maintenance records

Column Name	Data Type	Constraints	Description	Example
equipment_id	VARCHAR(10)	PRIMARY KEY	Unique equipment ID	EQ001
name	VARCHAR(100)	NOT NULL	Equipment name	Treadmill
category	VARCHAR(50)	NULL	Category	Cardio
status	ENUM('active','maintenance','inactive')	DEFAULT 'active'	Status	active
location	VARCHAR(50)	NULL	Location	Cardio Zone
last_maintenance	DATE	NULL	Last maintenance	2024-01-01
next_maintenance	DATE	NULL	Next maintenance	2024-02-01

SESSION_EQUIPMENT Table

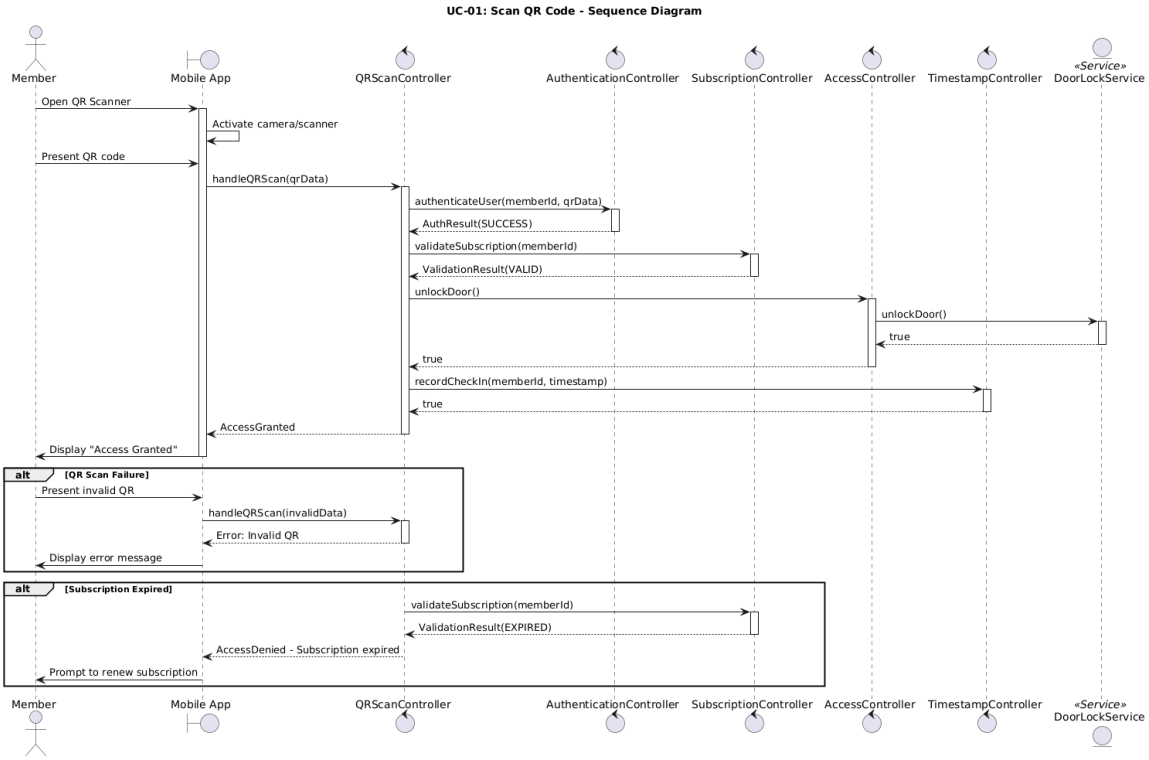
Description: Links workout sessions with equipment used.

Column Name	Data Type	Constraints	Description	Example
session_id	VARCHAR(20)	FOREIGN KEY, PRIMARY KEY	Session reference	SESS00120240115
equipment_id	VARCHAR(10)	FOREIGN KEY, PRIMARY KEY	Equipment reference	EQ001
duration_minutes	INT	NOT NULL	Duration of use	30
notes	VARCHAR(200)	NULL	Usage notes	Used for warm-up

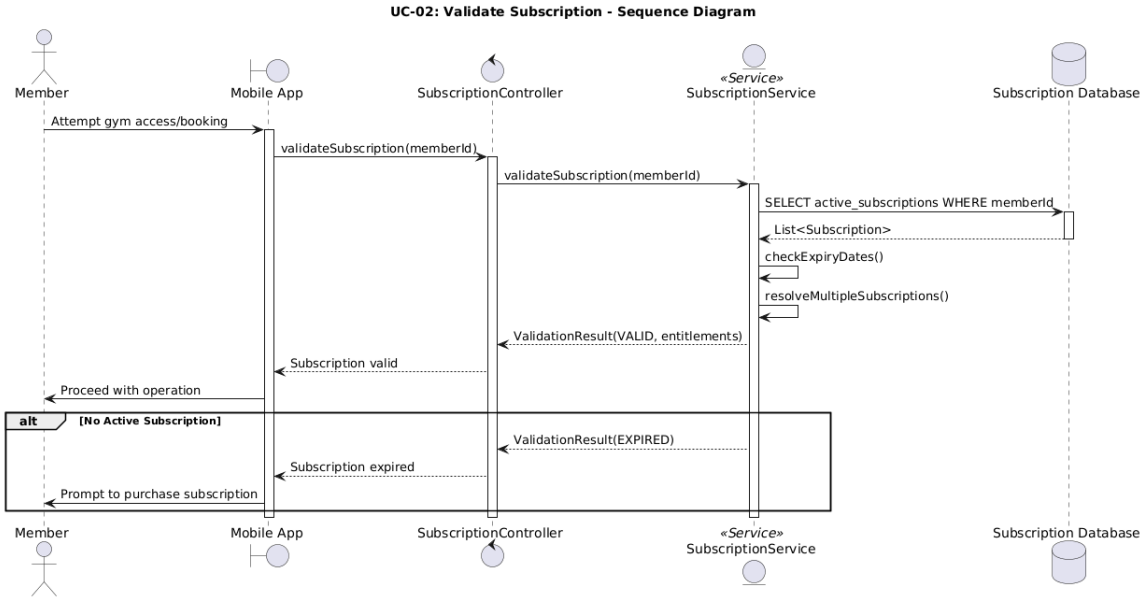
Sequence Diagrams

Sequence diagrams illustrate the step-by-step interactions between system components during a process. They show how messages are exchanged between actors and the system over time, making the flow of activities and system behavior easier to understand.

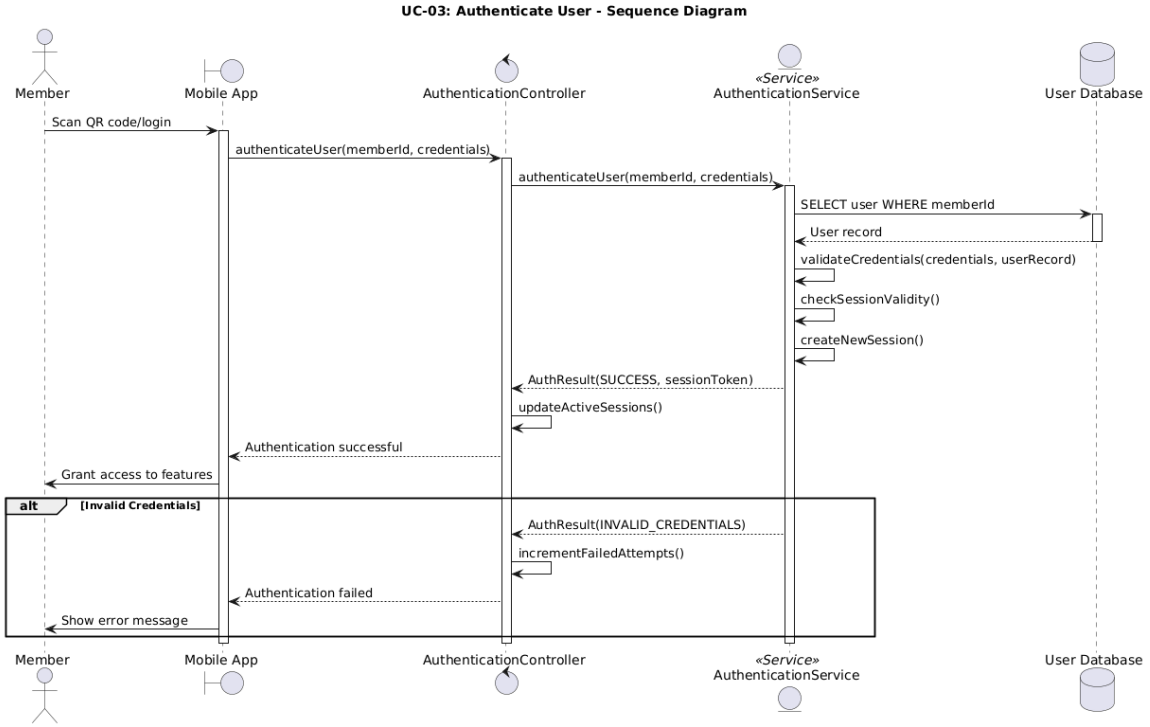
UC-01: Scan QR Code - Sequence Diagram



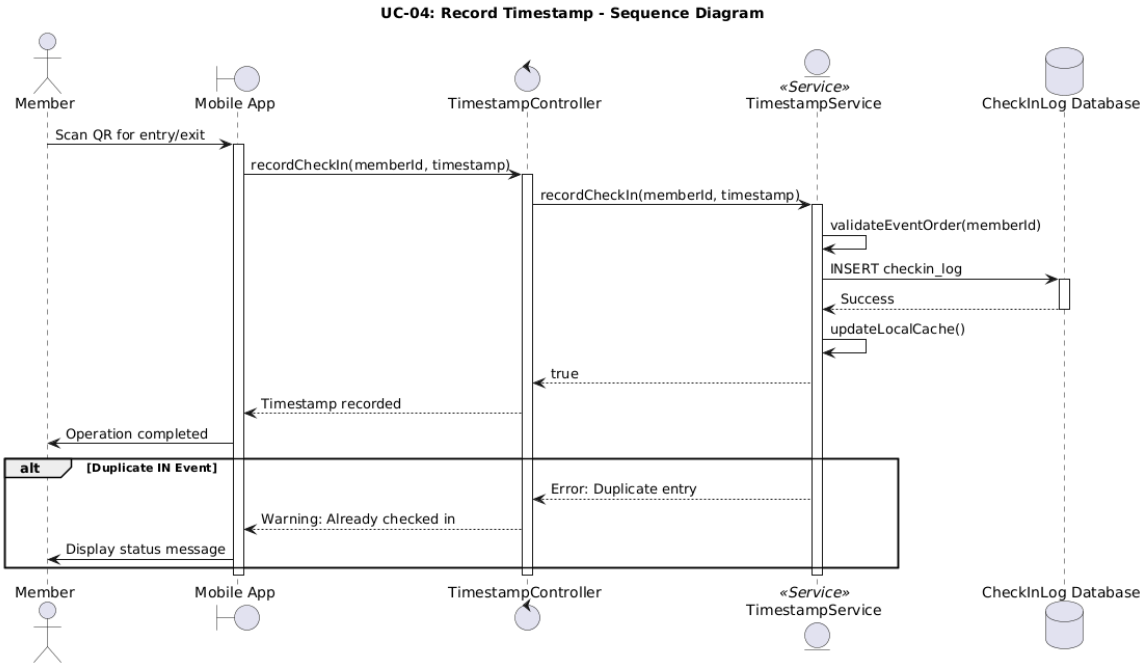
UC-02: Validate Subscription - Sequence Diagram



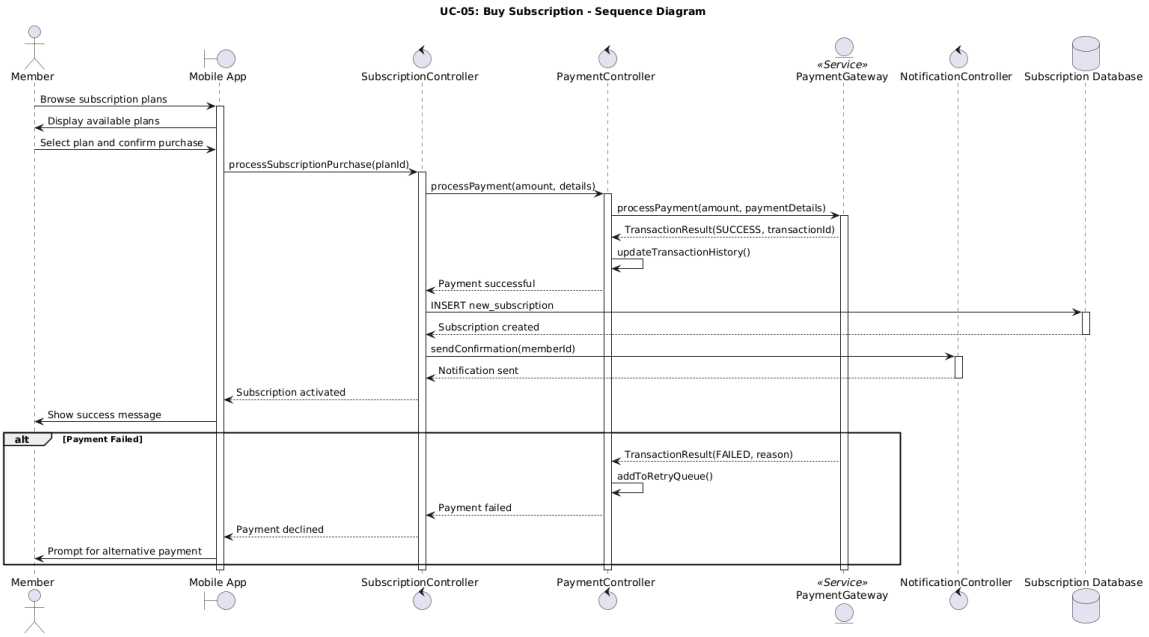
UC-03: Authenticate User - Sequence Diagram



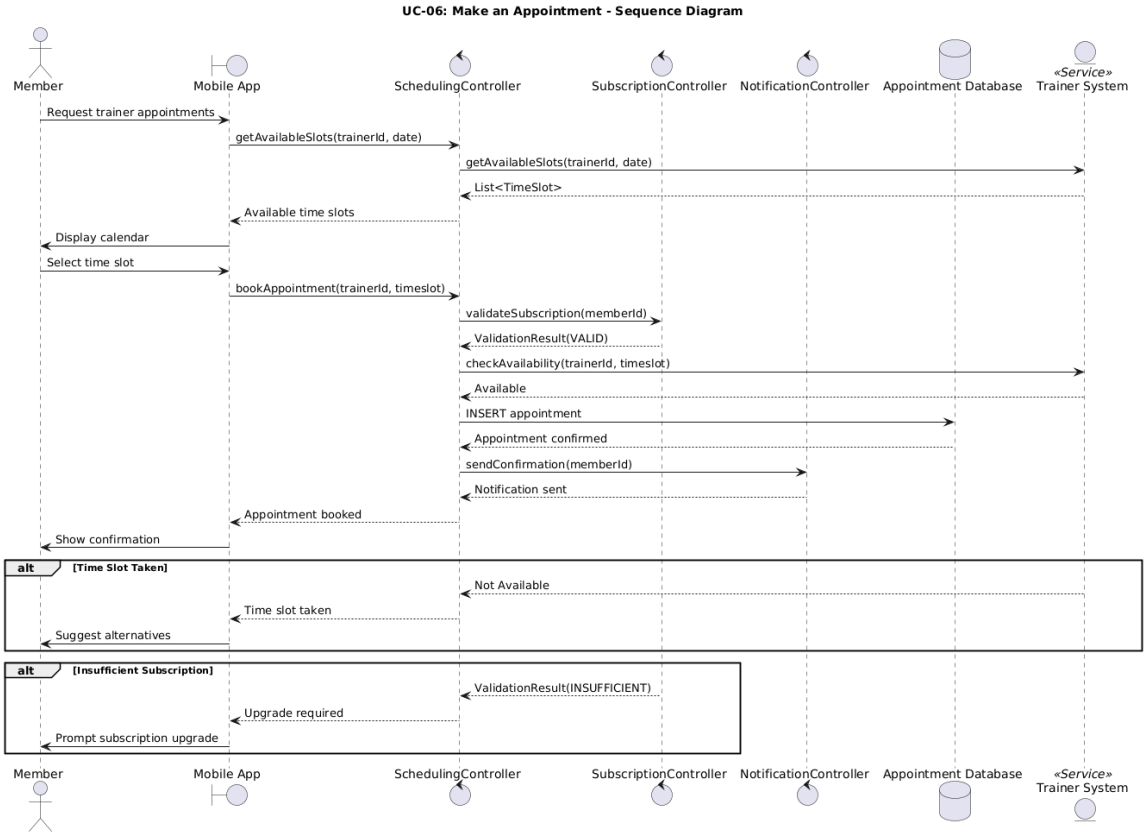
UC-04: Record Timestamp - Sequence Diagram



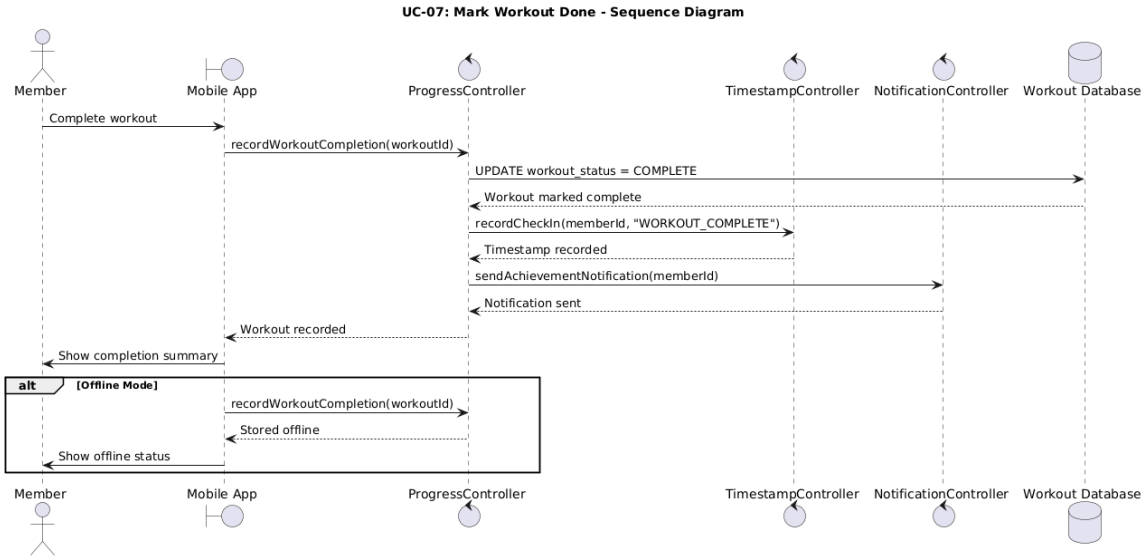
UC-05: Buy Subscription - Sequence Diagram



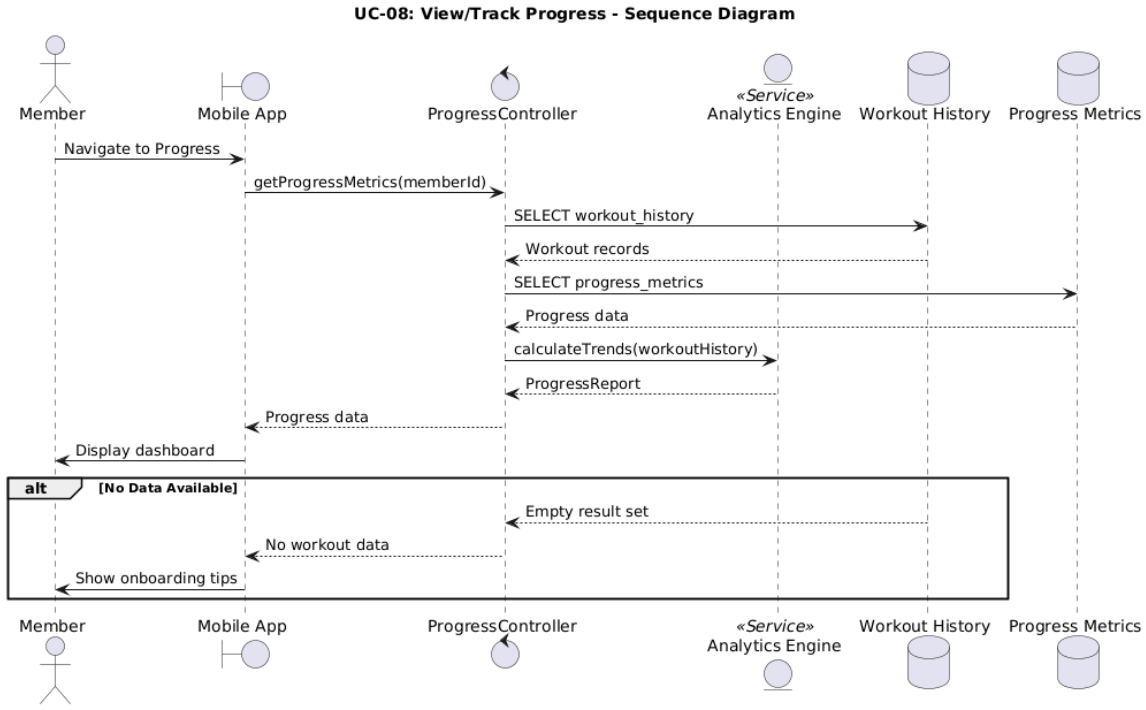
UC-06: Make an Appointment - Sequence Diagram



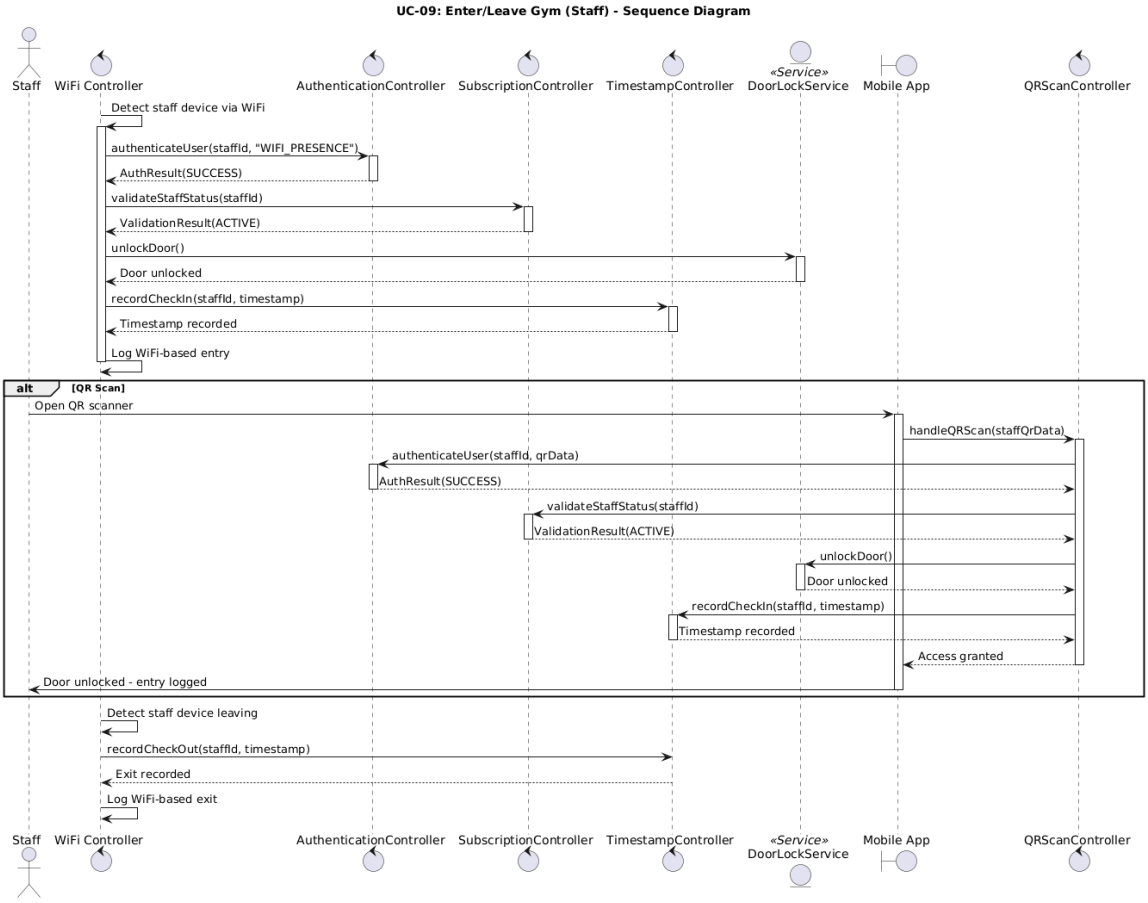
UC-07: Mark Workout Done - Sequence Diagram



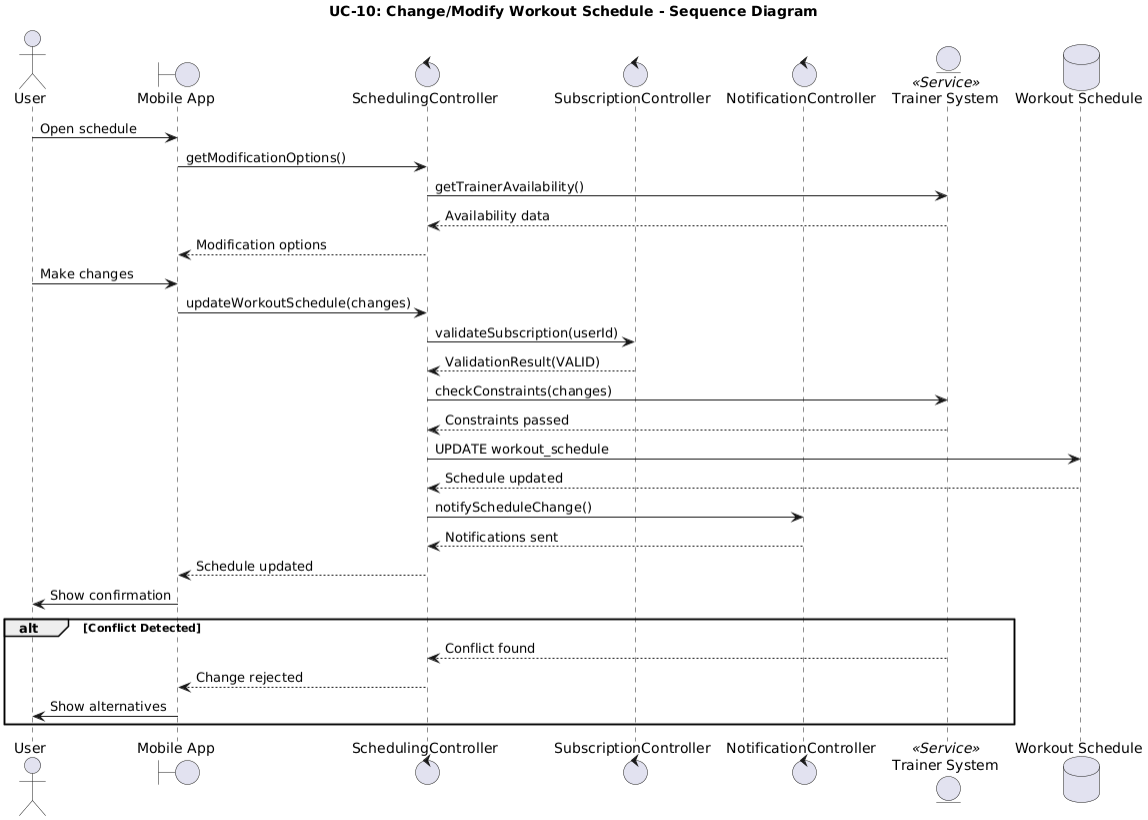
UC-08: View/Track Progress - Sequence Diagram



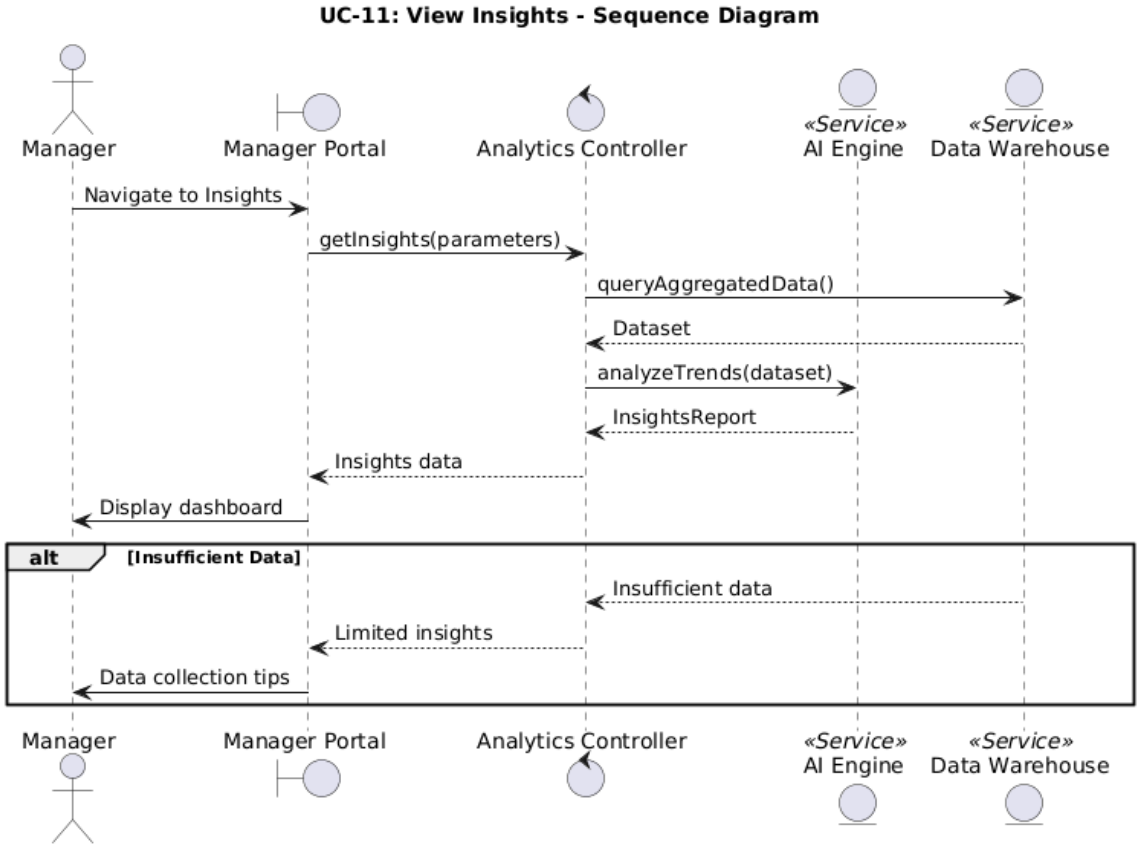
UC-09: Enter/Leave Gym (Staff) - Sequence Diagram



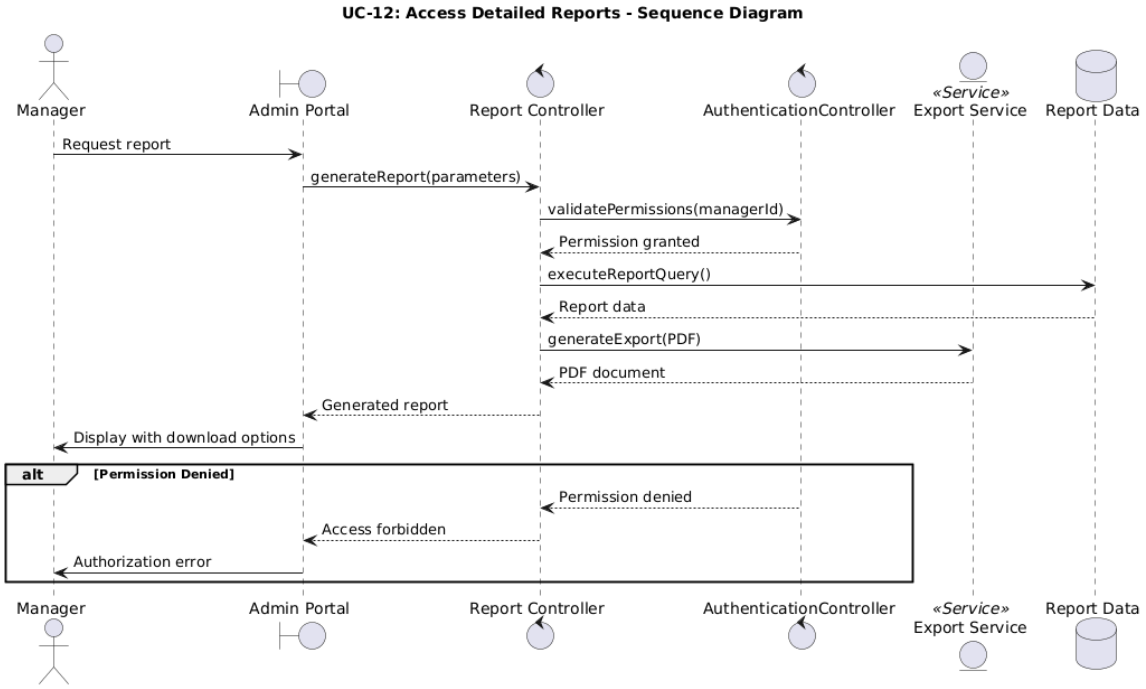
UC-10: Change/Modify Workout Schedule - Sequence Diagram



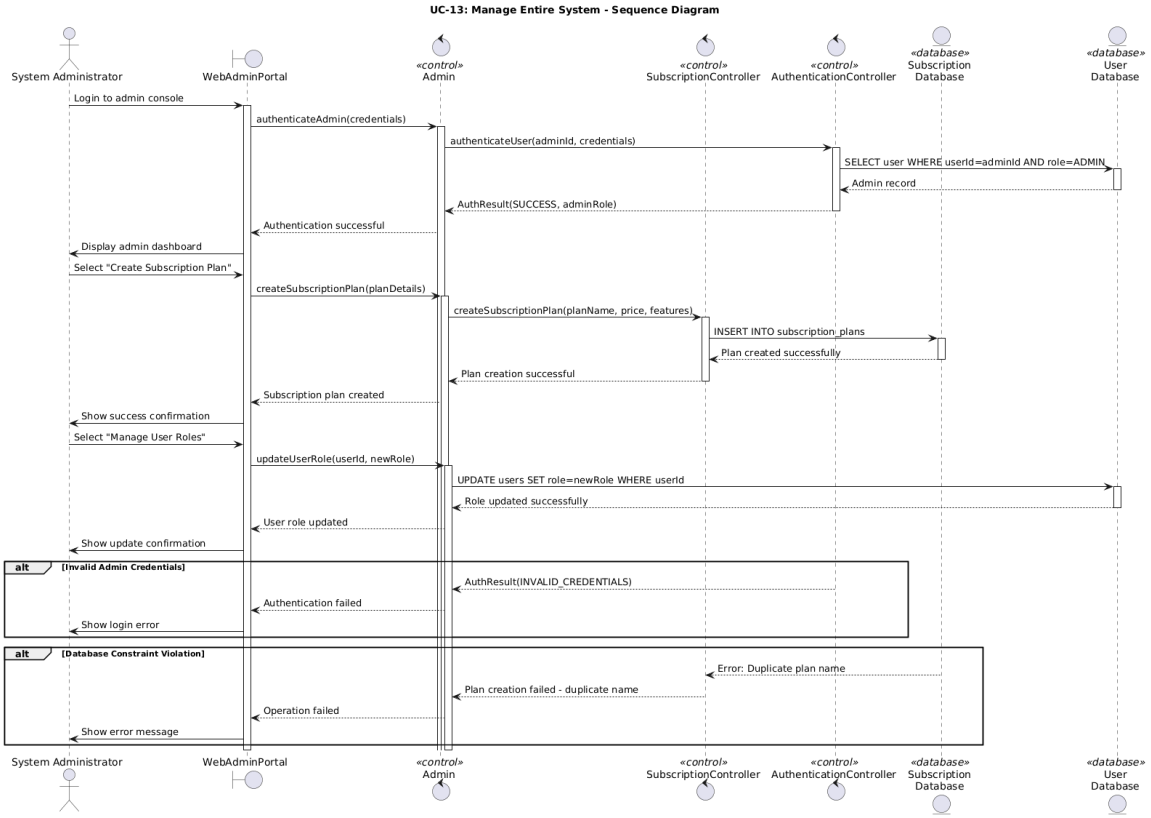
UC-11: View Insights - Sequence Diagram



UC-12: Access Detailed Reports - Sequence Diagram



UC-13: Manage Entire System - Sequence Diagram



***